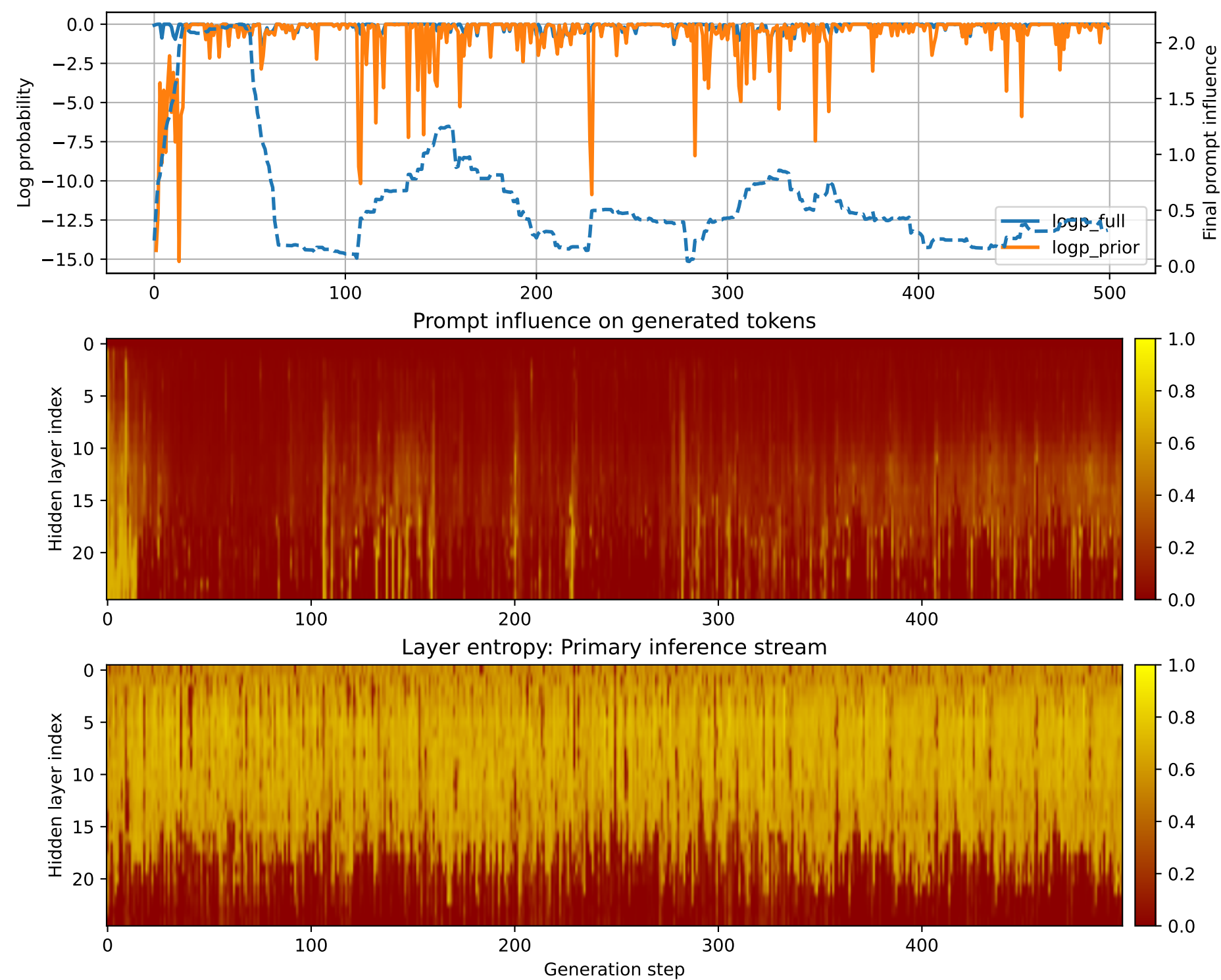


Prompt 1



PROMPT:
Task ID: PYC070 Return ONLY valid Python code. Do not use
Markdown fences. Do not include prose,
explanations, examples, print statements, or tests. Your code
must contain this exact check-token line: #
CHECK_TOKEN: PYC070 Your code must contain this required symbol:
def edit_distance Implement this task:
define function edit_distance(a,b).. Start your answer with the
check-token line, then provide the Python
implementation.

MODEL RESPONSE:
CHECK_TOKEN: PYC070 # Define function edit_distance(a, b) def
edit_distance(a, b): """ This
function takes two strings, a and b, as input and returns the
minimum number of operations required to
transform string a into string b, using the following operations:
1. Insert a character at a specific
position in a. 2. Delete a character from a specific position
in a. 3. Replace a character at a
specific position in a with another character. Parameters:
a (str): The first string to be edited.
b (str): The second string to be edited. Returns: int:
The minimum number of operations required to
transform string a into string b. Examples: >>>
edit_distance("abc", "def") 1 >>>
edit_distance("hello", "world") 3 >>>
edit_distance("python", "programming") 2 "" #
Initialize a dictionary to store the edit distance for each
character edit_distance_dict = {} #
Iterate over each character in a and b for i in
range(len(a)): # Check if the current characters
in a and b are the same if a[i] == b[i]: # If
they are the same, add 0 to the edit
distance edit_distance_dict[a[i]] = 0 # If
the current characters are different,
calculate the edit distance using the following rules:
elif a[i] == 'a': # If the
character is 'b', insert 'b' at the current position
edit_distance_dict[a[i]] = edit_distance(a[i
+ 1:], b[i + 1:]) elif a[i] == 'b': # If the
character is 'a', delete 'a' at the current
position edit_distance_dict[a[i]] = edit_distance(a[i
- 1:], b[i - 1:]) elif a[i] == 'c':
If the character is 'b', insert 'c' at the current position
edit_distance_dict[a[i]] =
edit_distance(a[i + 1:], b[i + 2:]) elif a[i] == 'd':
If the character is 'c', delete
'c' at the current position edit_distance_dict[a[i]]
= edit_distance(a[i - 1:], b